

Designing and Evaluating a Multimodal Transit Interface for First- and Last-Mile Decision-Making in Bangkok

Maethaphan Phaisalrattananukul*

Faculty of Engineering, Chulalongkorn University, maethaphan.phai@gmail.com, 6733211421@student.chula.ac.th

Napasorn Suwannakij

Faculty of Engineering, Chulalongkorn University, napasorn.swk@gmail.com, 6733118821@student.chula.ac.th

Penpitcha Piyawaranont

Faculty of Engineering, Chulalongkorn University, penpitcha.py@gmail.com, 6733185821@student.chula.ac.th

Assoc. Prof. Dr. Chutisant Kerdvibulvech

National Institute of Development Administration, chutisant.ker@nida.ac.th

Bangkok commuters often rely on multiple first- and last-mile connectors before reaching main transit, yet conventional route-planning interfaces usually present these segments as bundled route alternatives. This study evaluates whether explicitly visualizing and allowing selection of first-mile, main-transit, and last-mile components improves route-planning decisions. We conducted a within-subjects online experiment with 46 participants, comparing a baseline route-planning interface with an interactive web-application prototype First-Mile Last-Mile (FMLM) interface across realistic Bangkok travel scenarios. The prototype improved subjective ratings for decision ease (+0.396, $p = .005$) and FMLM understanding (+0.526, $p < .001$), and 29 of 46 participants rated FMLM information on the positive side of the usefulness scale. However, calculated preference-aligned route choice did not improve at the full-sample level (mean difference = -0.005, $p = .672$), suggesting a gap between perceived usefulness and measurable route-alignment outcomes. Qualitative feedback indicates that FMLM information is valuable, especially for unfamiliar routes, but current interface complexity, terminology, map encoding, and route-data trust limit its effectiveness. These findings highlight design requirements for FMLM-aware route-planning tools.

CCS CONCEPTS • Human-centered computing → Human computer interaction (HCI); Empirical studies in HCI; • Applied computing → Operations research; Transportation;

Keywords and Phrases: First-mile Last-mile, interface design, multimodal transit, transportation

1 INTRODUCTION

Bangkok has increasingly become a multimodal city, where commuters can combine several transport modes such as rail transit, buses, express boats, vans, motorcycle taxis, songthaews, ride-hailing, bike-sharing, and walking within a single trip. Shared and flexible mobility options are also relevant to Bangkok route choice, since prior work has examined the role of shared mobility in the city’s mode-choice landscape [2]. Even though major systems such as the Bangkok Mass Transit System (BTS) Skytrain, the Mass Rapid Transit (MRT) system, and express boat services provide relatively reliable transport, many commuters still face difficulties accessing these systems from their actual origins and destinations. This makes First- and Last-Mile (FMLM) connectivity a significant issue in Bangkok. Many trips require commuters to first reach certain station, pier, or some kind of main transit corridor, then continue from the stop to their desired destination. These access and egress segments may involve a combination of the aforementioned transport modes, where each option involves different trade-offs in travel time, cost, walking distance, reliability, safety, and convenience. As a result, commuters must often compare several imperfect alternatives before deciding how to complete a trip and may be one of the factors in the decline of public-to-private transport usage ratio [1]. This issue also exists within Bangkok’s broader car-oriented mobility context, where congestion and parking-related constraints have been discussed as structural factors shaping travel behavior [3].

Existing research and tools have addressed route planning through shortest-path recommendations, multimodal routing, accessibility visualization, and multi-criteria route ranking [29, 31, 36, 37]. Prior work also shows that public transport users need complete door-to-door information to reduce uncertainty [26, 27]. Nevertheless, existing route-planning interfaces often present complete route alternatives as bundled options, without clearly separating the first-mile, main-transit, and last-mile components. This makes it difficult for users to understand which part of the trip creates the main trade-off, or how changing one connector could affect the overall route. To address this, our study designs and evaluates an interactive web-application prototype route-planning interface for Bangkok by explicitly decomposing trips into first-mile, main-transit, and last-mile segments. This prototype allows users to inspect and select compatible access and egress connectors, rather than choosing only from complete pre-bundled routes. We compare this prototype with a baseline route-planning interface through a within-subjects online experiment with 46 participants across realistic Bangkok travel scenarios.

Thus, this study aims to make the following contribution:

1. To frame Bangkok FMLM route planning as a multi-attribute decision problem involving time, cost, walking distance, reliability, simplicity, and user preference.
2. To present a prototype interface that separates first-mile, main-transit, and last-mile choices for multimodal route-planning.
3. To evaluate the prototype against a baseline interface using subjective ratings, preference-aligned route-choice metrics, and qualitative feedback.
4. To identify design challenges for FMLM-aware route-planning tools, including terminology, visual complexity, and map encoding.

2 BACKGROUND, LITERATURE REVIEW, AND RELATED WORKS

This section reviews literature related to FMLM-aware route-planning interface design. First, it summarizes Bangkok’s multimodal transport context and the role of local connector modes. Second, it discusses commuter route-choice factors, including time, cost, walking distance, reliability, safety, comfort, and transfer burden. Third, it reviews information

uncertainty and cognitive load in public transport navigation. Finally, it discusses existing route-planning and transit-visualization approaches, identifying the gap addressed by this study.

2.1 Bangkok's Multimodal Transit and FMLM Context

Bangkok's transport system combines rail, road-based public transport, water-based transport, and local connector modes. Rail transit has expanded through BTS, MRT, and broader metro development plans, while buses, vans, and express boats continue to serve important travel corridors [4, 5, 6, 7, 8, 14]. However, the existence of multiple transport systems does not guarantee convenient door-to-door travel. Public transport use remains strongly affected by how easily commuters can access stations, piers, and main corridors from their actual origins and destinations. This access issue is especially important because Bangkok's urban form often separates residential areas from main public transport corridors. Many residential areas are located inside narrow residential roads, namely *sois*, while major transit services tend to operate along larger roads or fixed corridors [11]. As a result, commuters may need additional first-mile or last-mile modes before they can use rail, boat, or bus services effectively. For example, water-based transport can provide efficient corridor travel, but its use may be limited by long walking distances around piers [7].

Bangkok's FMLM environment includes both formal and less formal connector modes. Buses can operate as both main transit and feeder services, while vans and *songthaews* often provide local or corridor-based access to stations, activity centers, feeder areas, and commuting hubs [8, 9, 12, 13, 14]. Motorcycle taxis are also important because they can provide flexible short-distance access in congested areas and narrow *sois*, complementing rail transit in some areas and substituting for other public transport in others [10, 11]. Walking remains a necessary connector between modes, but willingness to walk depends on built-environment conditions, safety, comfort, and sidewalk quality [15, 16]. In addition, shared mobility and micromobility services, such as ride-hailing, bike-sharing, or scooter-sharing, can function as flexible FMLM connectors, although their usefulness depends on local policy, infrastructure, availability, and user adoption [34, 35]. These connector modes give commuters flexibility, but they also make route choice more complex. A route may be faster but more expensive, cheaper but require more walking, more direct but less reliable, or easier to understand but less efficiently. Therefore, Bangkok FMLM planning should not be treated only as a shortest-path problem. Instead, it requires support for comparing trade-offs between multiple route components and transport modes.

2.2 Bangkok Commuter's Decision-Making

The study in [17], which models within a Willingness to Pay space, found that walking is generally preferred over motorized modes such as *songthaews* and motorbike taxis, emphasizing its no-cost nature, simplicity, and flexibility. However, when given time-sensitive conditions, speed and reliability are valued more. Participants are also found to value safety and comfort too, since they are more willing to pay for helmet availability, improved walking paths, and seating in shared modes. [18] also confirms that walking is associated positively when safety, comfort, and sidewalk quality are met with commuters' expectations. For motorcycle taxis, however, users still feel driver-related concerns or discomfort from the lack of safety, resulting in suppressed use. For *songthaews*, service attributes, such as seating availability and service reliability, matter more to users than attitudinal preferences.

Not only do environmental factors matter, but as [1] found, age of commuters also influences the decision of choosing public transportation in the first place. The study in [1] found a significant decline in public transit use for commuters aged 30–49 and 50–65 compared to 2015 and 2023. This may suggest that as commuters age, lifestyle and income change how commuters perceive the need for public transportation, which is consistent with what [19] found in western societies. As discussed in [20], many factors, including ones already mentioned, influence mode choice behavior of commuters, such as

total time in public transport, total cost in private transport, and car ownership. However, this also depends on the traveler’s distance from the nearest public transport. For groups living more than 1 kilometer away from public transit, the same attributes were shown to affect mode choice behavior, except for travel time on public transport, suggesting that these commuters may prefer to use private vehicles instead.

Transfers also contribute to users’ decision-making, though there can be differences between intra-modal and inter-modal transfers. Depending on the mode and social context, results of general transfer as perceived extra transit time may not be the same, but it is found to be about 15.2–17.7 minutes of in-vehicle minutes equivalent, as mentioned in [21]. Yet in [22], the order in which the transfer to the mode occurs, to a bus in this case, also affects the perceived transfer penalty in waiting time. Another study in [23], which focuses on transfer disutility as Generalized Travel Time (GTT), found the optimal transfer time to be at 8 minutes, decreasing the GTT from 39–51 minutes to 29 minutes. Notably, factors surrounding the respondents and stations can affect the final time.

2.3 Challenges in Understanding FMLM Options

As we’ve established, there are many FMLM options available for commuters to connect to both rail and boat networks, and many of these options are sometimes informal. Other studies, such as [10, 13, 18], have highlighted the importance of these modes in connecting commuters to rail stations and boat piers, especially when road conditions are not ideal or do not allow larger vehicles to traverse [7]. Being as relevant as they are, however, representing their route information as predictably as other established transport modes remains a challenge.

One main challenge is that many FMLM options have not been documented to the same degree as formal transport options. Navigation systems typically depend on organized routes, stops, and scheduled data [24], but due to the nature of informal transit, this information may not be available to users in as much detail, including where to board, travel duration, cost, and reliability [25]. For example, while the songthaew dataset in [12] is available, some route data are outdated, which may lead to misleading information for some routes, such as route length and route path.

This lack of information can lead to perceived uncertainty. According to [26], commuters require information throughout the whole journey, from door to door, for them to feel comfortable and confident in reaching their destination. The lack of information for any part of the journey, such as where to transfer, where to board, or what connections are possible, can make users feel that public transportation is more difficult than it should be. Furthermore, [27] also mentions that trip information is needed most before the trip, while confirmation becomes more important as the trip continues. As a result, commuters who do not have full information may need to look up or search for the missing route information themselves. A study in [28] compares two route map schemes on cognitive load and found that while there is no significant difference in search efficiency, there is a difference in cognitive load between the two schemes; the map-based route map requires more cognitive load than the linear route map. The study further mentions that a highly symmetric linear route map can better reflect the structure of route information, allowing commuters to have a better grasp of the transit hierarchy.

Existing navigation tools, while often presenting route recommendations as alternatives, may not explicitly explain the trade-offs between each alternative. Especially in the case of FMLM travel, the “best” route for commuters can differ depending on what each commuter prioritizes the most [17, 18, 20]. Since public transit route information can influence commuters’ confidence and uncertainty [26], route choices should therefore be treated as a multi-attribute decision rather than a single shortest-path solution.

2.4 Existing Solutions and Related Works

Previous research has developed a few solutions closely related to the objective of our research and development in FMLM transit visualization. In [29], the paper develops a solution that identifies nearby FMLM stops that are worth showing to commuters based on multiple criteria, not just distance and shortest travel time. The paper proposes an optimization framework that calculates both route- and stop-based information. For route-based information, it considers total travel time, walking time, and number of transfers. On the other hand, for stop-based information, it considers service frequency and the number of bus lines serving the stop. The paper focuses on Pareto-optimal stops that are worth presenting and removes dominated stops, which are stops that are significantly worse on all attributes. These optimal stops result in measurable time savings, better frequency, significantly more bus lines, and fewer transfers than dominated stops. This implies that aspects other than distance and shortest time should be considered when planning a commute. Rather than only presenting one optimal route, the system can allow commuters to compare alternatives, which is also the focus of our paper but adjusted to fit the context of Bangkok transit.

Other related works focus on reducing confusion and uncertainty by improving the information available to commuters during multimodal travel. In [30], the paper developed a solution that aids people who are blind or have low vision in navigating public transit trips within the middle mile, or inside transit hubs. They created a mobile application that gives users real-time guidance in navigating subway stations, relying only on the user's smartphone camera, Optical Character Recognition (OCR), and transit data. While the solution may not be directly related to our research, it demonstrates that commuting trips can benefit from systems that reduce confusion and uncertainty, especially during multimodal transit, which is consistent with what is mentioned in [27]. A more relevant example can be found in [31], where the paper developed a web-based map for urban rail stations in North American cities, prioritizing real-time accessibility status. This study highlights the significance of the practicality of a transit option and puts a strong focus on mitigating missing information. Similarly, our study focuses on visualizing FMLM options to make trade-offs clearer and more understandable for commuters.

Another line of related work concerns how route alternatives are evaluated once multiple attributes are involved. This is important for the present study because the outcome is not defined by a single objectively correct route. Instead, the quality of a route choice depends on how closely the selected route matches what each participant says they value, such as shorter time, lower cost, less walking, higher reliability, or simpler trips. Personalized and multi-criteria route-planning studies commonly use the same basic evaluation logic as this study: route alternatives are described by several attributes, these attributes are normalized so they can be compared despite having different units, user or criterion weights are applied, and the weighted values are aggregated into a route score or impedance value. The study in [36], for example, presents a multi-criteria personalized route-planning model in which route-planning criteria are normalized, criteria importance reflects user preference, and the resulting criteria values are aggregated to calculate route impedance. Another study in [37] similarly proposes an ontology-based personalized route-planning system that combines route criteria, user preferences, and contextual information into a single impedance model for route selection.

Together, these studies support treating FMLM route selection as a preference-based, multi-attribute decision problem rather than a shortest-path problem alone. However, while prior work mainly focuses on generating, ranking, or supporting route recommendations, the present study focuses on how different interface presentations affect users' ability to understand, compare, and choose among FMLM route alternatives.

3 METHODOLOGY

3.1 System Design

The web application developed for this study consists of two interface conditions: a baseline interface and a prototype interface. The system was implemented as an interactive web application using Next.js 14.2.5 and React 18.3.1, with participant ID generation, local counterbalancing, route interaction, survey flow, and response logging integrated into the same platform [38, 39]. The live study was deployed on Vercel [40]. Route and response data were stored using Supabase, accessed through the Supabase JavaScript client version 2.105.4 [41]. All interfaces are in Thai, as the sample population consists of Thai participants. Overall, a total of 6 screen variants is developed, each of which will be shown throughout this section.

3.1.1 Interface Conditions

The two interfaces previously described are as follows.

First, the baseline condition, as shown in Figure 1 with in-image-text English translations in Table 1.

- Represents a conventional route-planning interface, like existing solutions
- Shows complete route alternatives
- Includes the map, total time, cost, walking distance, and modal icons.
- Specifically, does not let users separately choose between first- or last-mile options

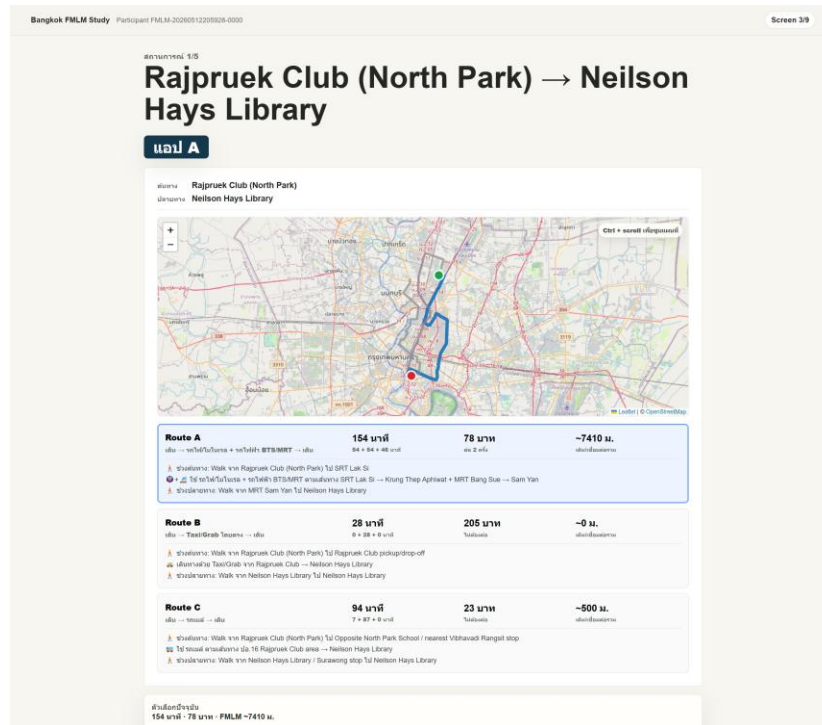


Figure 1: Baseline condition interface with map, total time, cost, and walking distance.

Table 1: English translation of in-image text within Figure 1.

Option	Details	Time	Cost	Walking distance
Route A — Walk + BTS/MRT + Walk	• Walk from Rajpruek Club (North Park) to SRT Lak Si • BTS/MRT: SRT Lak Si → Krung Thep Aphiwat → MRT Bang Sue → Sam Yan • Walk from MRT Sam Yan to Neilson Hays Library	154 min	78 ฿	7,410 m.
Route B — Walk + Taxi/Grab direct + Walk	• Walk from Rajpruek Club (North Park) to SRT Lak Si • BTS/MRT: SRT Lak Si → Krung Thep Aphiwat → MRT Bang Sue → Sam Yan • Walk from MRT Sam Yan to Neilson Hays Library	154 min	78 ฿	7,410 m.
Route C - Bus	• Walk from Rajpruek Club (North Park) to Opposite North Park School / nearest Vibhavadi Rangsit stop • Bus to Rajpruek Club area → Neilson Hays Library • Walk from Neilson Hays Library / Surawong stop to Neilson Hays Library	94 min	23 ฿	500 m.

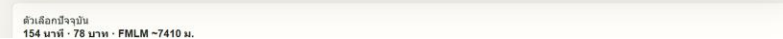
Note: Options and Details format in the table are close to what the participants are expected to see in the interface – only in Thai.

Second, the prototype condition, shown in Figure 2 with in-image English translations in Table 2

- Breaks trips into first mile, main transit, and last mile
- Let users choose the main route along with compatible first and last mile connectors
- Shows selectable connector options, color-coded route segments, and route summary cues like existing solutions, being fastest, cheapest, and recommended

Rajpruek Club (North Park) → Neilson Hays Library

ต้นทาง **Rajpruek Club (North Park)**
ปลายทาง **Neilson Hays Library**



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Table 2: English translation of in-image text within Figure 2.

Fastest ^a	Cheapest ^a	Fewest Transfers ^a		
28 min / 205฿	94 min / 23฿	28 min / 205฿		
Walk → Taxi/Grab direct → Walk ~0 m No transfer	Walk → Bus → Walk ~500 m No transfer	Walk → Taxi/Grab direct → Walk ~0 m No transfer		
Option ^b	Details ^b	Time ^b	Cost ^b	
Taxi/Grab direct (recommended)	Taxi/Grab from Rajpruek Club → Neilson Hays Library · ~25,650 m · No transfer	28 min	205฿	
Train + BTS/MRT	SRT Lak Si → Krung Thep Aphiwat + MRT Bang Sue → Sam Yan · ~28,390 m · 2 transfers	54 min	78฿	
Bus	Bus route 16 from Rajpruek Club area → Neilson Hays Library · ~21,620 m · No transfer	87 min	23฿	
Option ^c	Details ^c	Distance ^c	Time ^c	Cost ^c
Taxi/Grab (recommended)	Golf cart/internal shuttle then motorbike → SRT Lak Si	~2,300m	18 min	35฿
Golf cart/shuttle	Golf cart/internal shuttle toward public-road pickup → SRT Lak Si	~4,000m	29 min	Free
Walk	Walk from Rajpruek Club (North Park) to SRT Lak Si	~4,000m	54 min	Free
Option ^d	Details ^d	Distance ^d	Time ^d	Cost ^d
Taxi/Grab (recommended)	Motorbike taxi from MRT Sam Yan to Neilson Hays Library	~3,410m	16 min	41฿
MuvMi/Tuk-tuk	Tuk-tuk from MRT Sam Yan to Neilson Hays Library	~3,410m	23 min	150฿
Walk	Walk from MRT Sam Yan to Neilson Hays Library	~3,410m	46 min	Free

Note: Table 2 is divided into 4 different sections, denoted by separate column headers, since it is describing the same image. Section a describes the recommended route based on the fastest, cheapest, and shortest connection. Section b describes main-route options. Section c describes first-mile options. Section d describes last-mile options.

3.1.2 Route Data and Map Representation

All study data within the web application is stored via the Supabase database. The 10 study scenarios were derived from the pre-study survey responses, which provided origin-destination patterns and participant travel preferences. For each scenario, the researchers then compiled route options and route attributes using available public transport information, manual route checking, and manually prepared Global Positioning System (GPS) Exchange Format (GPX) route geometries. The database includes 10 scenarios, 35 main routes, 98 first-mile options, 84 last-mile options, and 182 FMLM option rows. We also keep route attributes such as travel time, cost, distance, mode, concise participant-facing detail text, reliability score, ranking values for time, cost, distance, reliability, simplicity, and scoring metrics used for evaluation. The criteria used to select the 10 study scenarios are described in Section 3.2.3, “Pre-study survey and scenario selection.”

Maps presented in the system use OpenStreetMap basemap tiles [42] rendered through Leaflet 1.9.4 and React Leaflet 4.2.1 [43, 44]. Actual route paths are overlaid using GPX files, while origin and destination markers are stored with each scenario. Both main routes and FMLM connector routes are also represented with GPX geometries as available.

3.2 Experiment Design

This study’s objective is to evaluate whether explicitly presenting and allowing users to choose first- and last-mile access options improve commuter decision-making in Bangkok.

3.2.1 Study Design

We employ a comparative user study with a within-subjects design. As mentioned, the system contains two kinds of interfaces, and both interfaces will be used by every participant along the system flow. This is to enable direct individual-level comparison while reducing variance from differences in participants’ own socioeconomic background. Notably, both interfaces include a map as the background for route display to reduce the influence of having or not having a map [28], and put the focus more towards the granularity, the ability to have choices, and structure of FMLM information. Furthermore, both condition order was counterbalanced across participants to control ordering and learning effects that may occur. The interface has clear labels toward the participants as App A and App B, but not directly telling which interface is in which condition.

3.2.2 Participants

Participants are recruited from individuals who commute within Bangkok or are familiar with the city. We targeted diverse travel frequency, transport modes used, navigation app experience, and familiarity with multimodal travel.

The target sample was 40 complete submissions. This target was selected as a practical minimum for an online within-subjects interface experiment rather than as a population-representative survey sample. A within-subjects design was appropriate because each participant completed both interface conditions, allowing paired comparisons within the same participant and reducing between-participant variability compared with a between-subjects design [45]. For planning purposes, we used the standard paired-difference power approximation:

$$n \approx \left((Z_{1-\alpha/2} + Z_{1-\beta}) / d_z \right)^2 \quad (1)$$

where α is the significance threshold, $1-\beta$ is statistical power, and d_z is the standardized mean within-participant difference. The effect size d_z was defined as μ_d / σ_d , following Lakens’ notation for paired-sample designs [47]. With $\alpha = .05$ and target power = .80, a target of 40 participants is sufficient to detect an effect of approximately $d_z \approx 0.45$ – 0.46 , close to Cohen’s conventional medium effect size. This target is also consistent with Human-Computer Interaction guidance that sample size should be interpreted relative to study method and evidence goals [48], rather than treated as a fixed universal number.

The final dataset contained 46 complete submissions before the survey was closed, exceeding the target. Each completed submission included 5 assigned scenarios completed under both interface conditions. Therefore, the dataset contained $46 \times 5 = 230$ paired scenario-level comparisons between baseline and prototype, where each comparison paired the baseline and prototype results from the same participant on the same assigned scenario.

Participation is voluntary and anonymous. Our study does not collect any personal identifiable information. Participants may stop at any time. Partial interaction data may be logged during the study, but only completed responses are used for the primary analysis. The study does not claim that 40 complete submissions statistically represent all Bangkok travel behavior. Instead, diversity of travel patterns was addressed through the scenario selection and task allocation procedures, described in Sections 3.2.3 and 3.2.5.

3.2.3 Pre-Study Survey and Scenario Selection

To derive realistic route scenarios, we also conducted a pre-study survey where participants reported their origin and destination sub-district and district, the transport mode they use, and their priorities in selecting routes as can be seen in Figure 3. It was conducted as an online questionnaire. The respondents include Bangkok-based working parents, various university students, and Facebook community members who use all types of buses and trains. Both groups are large, with over 300K followers on their pages.

Regarding data collection, the pre-survey was conducted from April 24 to April 28, 2026. The main survey (Bangkok FMLM Study) was also conducted as an online questionnaire. More than 80% of the respondents in the main survey had previously participated in the pre-survey, which helped ensure faster survey completion by May 12, 2026. The survey collected a total of 111 responses across 60 unique origins and 53 unique destinations. An example of the responses is shown in Figure 4. Priorities included in the survey are speed, simplicity, cost, reliability, and walking distance. Travel modes included in the survey are private vehicles, urban rail (metro), walking, buses, and other semi-formal/informal FMLM modes.

Figure 3 displays the pre-study survey interface, which consists of four main sections:

- Question 1: คุณอยู่แถวไหน? Where do you live?** This section asks for the origin location, with dropdown menus for District and Subdistrict.
- Question 2: คุณมักจะเดินทางไปแถวไหน? Where do you often go?** This section asks for the destination location, with dropdown menus for District and Subdistrict.
- Question 3: How do you usually get there? คุณเดินทางไปที่นั่นด้วยวิธีใดบ้าง? (วิธีไหนบ้างที่คุณเคยใช้บ้าง)** This section lists various transport modes with corresponding icons: Walk, Bus, Van, Songthaew, Taxi, Motorcycle, Bicycle, Own, Urban Rail, Train, and Boat.
- Question 4: The top priority when deciding on a travel route? สิ่งสำคัญที่สุดเมื่อเลือกเส้นทาง (เลือก 1 ข้อ)** This section lists five priorities with radio button selection: Fastest, Cheapest, Least walking, Most reliable, and Easiest / Least confusing.

Figure 3: Pre-study survey

Route selection followed a two-stage process. In the first stage, all 111 records from the pre-study survey were spatially visualized as district- and subdistrict-level desire lines based on the reported origin-destination areas to identify overall travel patterns. The visualization as shown in Figure 5 revealed that most trips converged toward central Bangkok, particularly Pathum Wan, Wang Mai, Silom, and Sathon districts, while origins were distributed across all directions, including peripheral areas such as Don Mueang, Nong Chok, Lat Krabang, and Bang Khun Thian. Several outlier routes were also identified, connecting remote suburban areas to the city center over distances exceeding 20 km.

In the second stage, 10 representative Bangkok origin-destination pairs, or scenarios, were selected to represent varied geographic corridors and different FMLM challenges. The selection was based on three criteria: (1) spatial coverage across all four cardinal directions of Bangkok, (2) diversity of transit modes involved, and (3) the severity of FMLM access challenges. Origins cover most areas around Bangkok, such as northern, eastern, western, and southern Bangkok, representing residential areas or common places that the population generally commutes from. Destinations cover central business district (CBD) areas, institutional areas, and medical areas such as Silom/Sathon, Asok, Siam/Chula, Chatuchak, Phra Nakhon, Ekkamai, and Siriraj. We specifically selected scenarios where FMLM access is non-trivial, such as routes starting deep inside sois, areas with difficulty reaching metro stations, river-separated trips, large institutional sites, or destinations requiring local connector travel. As exact coordinates of individual respondents were not collected to protect participant privacy, representative origin and destination sites within each identified corridor were subsequently

approximated using subdistrict centroids and researcher-guided AI-assisted site selection, with FMLM accessibility challenge serving as the primary criterion. Therefore, this approach constitutes purposive sampling rather than statistical sampling.

id	home_district	home_subdistrict	dest_district	dest_subdistrict	transport	route_priority
1	หลักสี่	ทุ่งสองห้อง	ปทุมวัน	วังใหม่	["walk","bts","bus"]	fastest
2	ดินแดง	ดินแดง	ปทุมวัน	ปทุมวัน	["mrt"]	reliable
4	ดอนเมือง	สีกัน	ปทุมวัน	ปทุมวัน	["bts","mrt"]	cheapest
5	คลองสาน	บางลำภูล่าง	ปทุมวัน	ปทุมวัน	["bus","boat","walk"]	cheapest
6	ปทุมวัน	วังใหม่	ปทุมวัน	วังใหม่	["bus","walk","moto"]	fastest
7	บางนา	บางนา	ปทุมวัน	วังใหม่	["walk","car","bts","mrt"]	fastest
9	ปทุมวัน	วังใหม่	ปทุมวัน	วังใหม่	["walk","bus"]	fastest
10	บางรัก	บางรัก	บางรัก	บางรัก	["walk","bus","moto"]	easiest
11	บางพลัด	บางพลัด	ปทุมวัน	วังใหม่	["mrt","moto"]	reliable
13	ปทุมวัน	รองเมือง	ธนบุรี	บุคคโล	["bus"]	reliable
14	้วยขวาง	บางกะปิ	พญาไท	พญาไท	["mrt"]	easiest
15	บางพลัด	บางพลัด	บางรัก	สุริยวงศ์	["own"]	easiest
16	พญาไท	บางซื่อ	บางซื่อ	บางซื่อ	["walk","urban_rail"]	easiest
17	ดินแดง	รัชดาภิเษก	ปทุมวัน	ปทุมวัน	["urban_rail","moto","walk"]	fastest
18	หลักสี่	ทุ่งสองห้อง	ปทุมวัน	ปทุมวัน	["urban_rail","own"]	easiest
19	บางซื่อ	บางซื่อ	ลาดพร้าว	ลาดพร้าว	["urban_rail","taxi","own"]	easiest
20	บางซื่อ	บางซื่อ	ปทุมวัน	วังใหม่	["own","urban_rail"]	fastest
21	พระนคร	วัดราชบพิธ	ปทุมวัน	วังใหม่	["urban_rail"]	fastest
22	บางซื่อ	บางซื่อ	ปทุมวัน	วังใหม่	["urban_rail","walk"]	easiest
23	บางกะปิ	คลองจั่น	วัฒนา	คลองเตยเหนือ	["own"]	easiest
24	คลองเตย	พระโขนง	ปทุมวัน	ปทุมวัน	["urban_rail","own"]	fastest
26	บางเขน	อนุสาวรีย์	ปทุมวัน	ปทุมวัน	["urban_rail"]	fastest
27	สาทร	ช่องนนทรี	วัฒนา	คลองตันเหนือ	["walk","urban_rail"]	easiest
28	คลองเตย	คลองเตย	ปทุมวัน	ปทุมวัน	["urban_rail","bus","moto","own"]	fastest

Figure 4: Example of pre-survey responses. Note that district and sub-district data shown are in Thai.

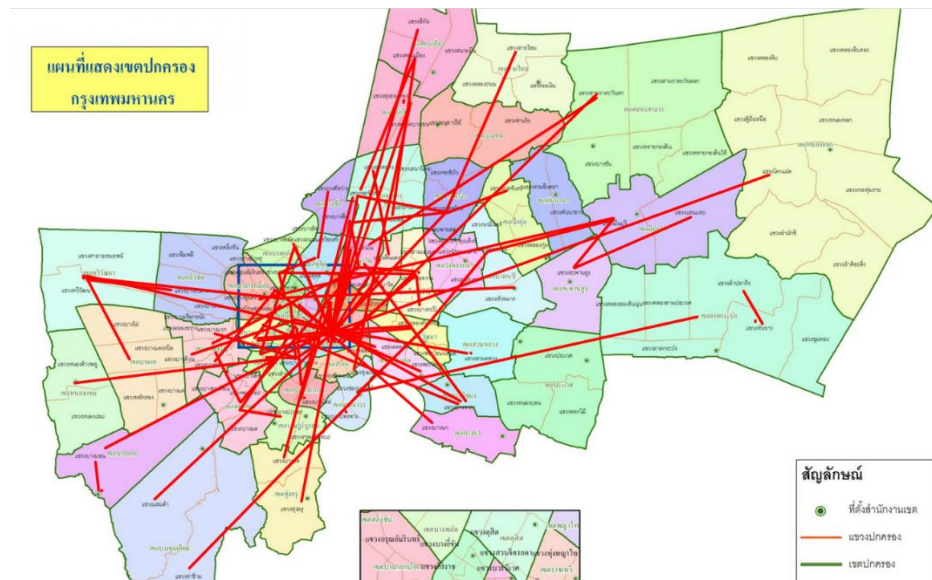


Figure 5: Origin-destination visualization from pre-survey responses. The map used shows the administrative boundaries of Bangkok.

The legend denotes from top to bottom as follows: District office location, sub-district administrative boundaries, and district administrative boundaries.

3.2.4 The 10 Selected Scenarios

The final 10 scenarios that were selected are as follows:

- Lak Si to Silom: Rajpruek Club (North Park) to Neilson Hays Library
- Bang Na to Sathon: Bangkok Patana School to Saint Louis College
- Min Buri to Asok: Ruamrudee International School to Ocean Tower 2
- Bang Wa to Chula/Siam: Khlong Bang Luang Floating Market to Chulalongkorn University Centenary Park
- Don Mueang to Chatuchak: Happy Avenue Don Mueang to Queen Sirikit Park
- Phra Khanong to Grand Palace: Kasem Phitthaya School to The Grand Palace
- Khlong San to Giant Swing: Lhong 1919 to Bangkok City Hall
- Bang Phlat to Ari: Wat Awut Wikasitaram to Ministry of Finance
- Suan Luang to Ekkamai: Triam Udom Suksa Pattanakarn School to Donki Mall Thonglor-Ekkamai
- Phasi Charoen to Siriraj: Wat Paknam Bhasicharoen to Wang Lang Market

Each scenario contains multiple main route alternatives. Although the initial design targeted 3 main routes per scenario, the final database includes all available verified routes from the route sheet. Main routes differ by transit strategy, such as rail-based routes, bus routes, boat routes, road-based routes, or multimodal combinations. Each main route is associated with compatible first-mile and last-mile options. These include walking, bus, songthaew, boat connectors, taxi/Grab, GrabBike, and other locally relevant connectors where appropriate. Walk options are always included as a baseline connector, but the prototype intentionally lets participants replace long or inconvenient walks with feasible non-walk FMLM modes when available. Additional non-walk options were added for routes where walking exceeded approximately 1.5 kilometers, and a plausible alternative was available.

Table 3: The 10 selected routes, along with their exact origin-destination estimation and FMLM challenge associated with them.

#	Corridor	Origin Area	Estimate Origin	Destination Area	Estimate Destination	FMLM Challenge
1	North	Lak Si	Rajpruek Club (North Park)	Silom	Neilson Hays Library	Origin is deep inside North Park estate; route uses SRT Lak Si → Krung Thep Aphiwat + MRT Bang Sue → Sam Yan, or Bus 16; destination requires walking into a soi off Silom road
2	East	Bang Na (Soi Lasalle)	Bangkok Patana School	Sathon	Saint Louis College	School is 3+ km deep in Soi Lasalle with no direct bus access; route uses BTS Bearing → Saint Louis, or Van 67 from Bang Na expressway underpass
3	East	Min Buri (private soi)	Ruamrudee International School	Asok	Ocean Tower 2	Origin is in a private soi with no public transit access; route uses Pink Line Min Buri → BTS Asok, or Bus 549/S4 to Lat Krabang + ARL Makkasan

4	West	Bang Wa	Khlong Luang Floating Market	Bang Floating	Chula / Siam	CU Centenary Park	Market requires 15–20 min walk through narrow alleys to reach; route uses MRT Bang Phai → Sam Yan, or BTS Silom Line Bang Wa → National Stadium
5	North	Don Mueang	Happy Don Mueang	Avenue Don Mueang	Chatuchak	Queen Sirikit Park	Limited public transit on Song Prapha road; route uses SRT Don Mueang → Bang Sue + MRT Chatuchak Park, or Bus 510/59; 20 min walk to reach the park center
6	East	Phra Khanong soi	Kasem (sub-Phitthaya School soi)		Phra Nakhon	The Grand Palace	Origin is deep in a sub-soi; route uses BTS Phra Khanong → Saphan Taksin, BTS Ekkamai → Saphan Taksin, or BTS Phra Khanong → Asok + MRT Sanam Chai; 15–20 min walk along palace perimeter walls to entrance
7	West	Khlong San (riverside)	San Lhong 1919		Giant Swing area	Bangkok City Hall	Origin isolated by Chao Phraya river; route uses cross-river ferry Tha Din Daeng → Ratchawong Pier, Orange Flag ICONSIAM → Tha Chang, or Bus 43 EV; City Hall requires 10–15 min walk from MRT Sam Yot
8	West	Bang Phlat (riverside)	Wat Wikasitaram	Awut	Ari area	Ministry of Finance	Temple is by the river with limited road access; route uses MRT Bang Phlat/Sirindhorn → Kamphaeng Phet, or Orange Flag Sang Hi Pier → Thewet Pier; Ministry of Finance requires 20–25 min walk from the pier
9	East	Suan Luang (large campus)	Triam Udom Suksa Pattanakarn School		Ekkamai	Donki Mall Thonglor–Ekkamai	Large school campus with dispersed exits; route uses ARL Hua Mak → BTS Thong Lo/Ekkamai via Makkasan, or BTS On Nut → Thong Lo/Ekkamai; mall requires 15–20 min walk from BTS Ekkamai
10	West	Phasi Charoen	Wat Bhasicharoen	Paknam	Siriraj	Wang Lang Market	Requires 15–20 min walk through narrow alleys from MRT to temple; route uses MRT Bang Phai/Bang Wa → Itsaraphap, or Khlong Phasi Charoen boat to Taksin-Phet

3.2.5 Study Procedure

All participants followed this study flow through the web application.

1. Consent screen, as shown in Figure 6. Participants are given a Thai-language consent page explaining the study purpose, expected task, estimated duration, voluntary participation, and privacy notice. They must accept the terms and conditions before continuing the survey.

แบบสำรวจประกอบการวิจัย การเข้าถึงระบบขนส่งช่วงต้นทางและปลายทางในกรุงเทพฯ

คำแนะนำสำคัญ
ในแต่ละสถานการณ์ กรุณาเลือกเส้นทางที่คุณอยากใช้มากที่สุดตามความต้องการของคุณเอง ไม่มีคำตอบที่ถูกต้องหรือผิด

วัตถุประสงค์
การศึกษานี้ต้องการเปรียบเทียบวิธีนำเสนอข้อมูลเส้นทาง เพื่อทำความเข้าใจว่าข้อมูลการเดินทางช่วงต้นทางและปลายทาง (FMLM) ช่วยให้ผู้ตัดสินใจเลือกเส้นทางได้ดีขึ้นหรือไม่

สิ่งที่ผู้เข้าร่วมจะทำ
คุณจะ:
• ตอบคำถามพื้นฐานเกี่ยวกับรูปแบบการเดินทางของคุณ
• ทดลองเลือกเส้นทางจากสถานการณ์การเดินทางหลายสถานการณ์
• เปรียบเทียบหน้าจอแอป 2 รูปแบบ ได้แก่ แอป A และแอป B
• ให้คะแนนความเข้าใจ ความง่ายในการตัดสินใจ และความพึงพอใจ

ระยะเวลา
ใช้เวลาประมาณ 10-15 นาที

ความสมัครใจ
การเข้าร่วมเป็นไปโดยสมัครใจ คุณสามารถหยุดทำแบบทดลองได้ทุกเมื่อ โดยไม่มีผลเสียใดๆ

ข้อมูลและความเป็นส่วนตัว
ระบบจะไม่ขอข้อมูลส่วนตัวของคุณ ข้อมูลที่เก็บรวบรวมถึงค่าตอบของคุณ เส้นทางที่เลือก คะแนนเปรียบเทียบ และข้อมูลการใช้งาน เช่น ลำดับหน้าจอหรือเวลาที่ใช้ตอบ ข้อมูลจะไว้เพื่อการวิจัยเท่านั้น

☐ ฉันได้อ่านข้อมูลข้างต้น เข้าใจ และยินยอมเข้าร่วมการศึกษา

Bangkok FMLM Study — Research Consent Form First-Mile Last-Mile Transportation Access in Bangkok

- **Purpose** To compare how route information is presented, in order to understand whether FMLM (first-mile/last-mile) data helps users make better route decisions.
- **What You'll Do**
 - Answer background questions about your travel habits
 - Try selecting routes from various simulated travel scenarios
 - Compare 2 app interfaces (e.g., App A vs App B)
 - Rate comprehension, ease of decision-making, and satisfaction

Time Commitment Approximately 10–15 minutes

Voluntary Participation Fully voluntary — you may stop at any time with no consequences

Privacy No personal identifying information collected. Data gathered includes: answer sequences, chosen routes, ratings, and usage patterns (e.g., screen order, response time). Used for research purposes only.

Figure 6: Screen 1 of the system, asking for participant's consent

2. Background and preference questionnaire screen, as shown in Figure 7. Participants answer questions on travel frequency, transport modes used, navigation app usage, route preference ranking, and familiarity with multimodal travel involving several connector stages.

Bangkok FMLM Study
Participant FMLM-20260512205528-0000
Screen 2/9

ข้อมูลก่อนเริ่มสถานการณ์
ข้อมูลพื้นฐาน + ความชอบ
Background + Preferences Questionnaire

Q1: คุณเดินทางบ่อยแค่ไหน
How often do you travel?
☐ ทุกวัน
Every day
☐ 3-5 วันต่อสัปดาห์
3-5 days/week
☐ 1-2 วันต่อสัปดาห์
1-2 days/week
☐ นาน ๆ ครั้ง
Rarely

Q2: รูปแบบการเดินทางที่ใช้
Transportation modes you use

เดิน
Walk

รถเมล์ไปประจำทาง
Bus

รถตู้
van

รถสองแถว
Songthaew

แท็กซี่
Taxi

มอเตอร์ไซด์
Motorbike

จักรยาน(เช่า)
Bicycle (rental)

รถยนต์-จักรยาน-มอเตอร์ไซด์ส่วนตัว
Personal vehicles

รถไฟฟ้า(ทุกสาย)
Urban Rail

เรือ
Boat

Q3: คุณใช้แอปนำทางบ่อยแค่ไหน
How often do you use navigation apps?
☐ บ่อยมาก
Often
☐ บางครั้ง
Sometimes
☐ นาน ๆ ครั้ง
Rarely
☐ ไม่เคยใช้
Never

Q4: ลากเพื่อจัดอันดับปัจจัยที่สำคัญที่สุด
Rank your current travel priorities (drag to reorder)

1

🕒

ใช้เวลาน้อยที่สุด

Least time

⬆️⬆️

2

💰

ค่าใช้จ่ายต่ำที่สุด

Lowest cost

⬆️⬆️

3

🚶

เดินน้อยที่สุด

Least walking

⬆️⬆️

4

🕒

เชื่อถือได้ว่าเวลาค่อนข้างแน่นอน

Reliable/consistent travel time

⬆️⬆️

5

👉

เข้าใจง่าย/ไม่ซับซ้อน

Easy to understand/not complicated

⬆️⬆️

Q5: คุณคุ้นเคยกับการเดินทางที่ต้องต่อหลายช่วงจนถึงสามหลัก เช่น เดิน → รัน → สองแถว → รถไฟฟ้า → เดินหรือต่อรถ อีกครั้ง มากแค่ไหน

1

ไม่คุ้นเคยเลย

2

ไม่ค่อยคุ้นเคย

3

ปานกลาง

4

ค่อนข้างคุ้นเคย

5

คุ้นเคยมาก

ไปฝั่งสถานการณ์

Q5: How comfortable are you with multi-leg journeys? (e.g. Walk → Bus → Songthaew → BTS → Walk again) Scale 1-5: Not comfortable at all → Very comfortable.

Figure 7: Screen 2 of the system, asking general transit questions similar to pre-study survey

- Scenario tasks, as shown in Figure 8. Each participant was assigned 5 of the 10 scenarios using a seeded shuffle based on participant ID to distribute participant exposure across the scenario pool, all 10 scenarios received participant feedback in the final analytical dataset, with coverage ranging from 17 to 31 participants per scenario. The same 5 scenarios are completed under both interface conditions. Figures for each scenario are similar to Figure 1 and 2.
- Interface exposure. For each scenario, participants use one interface, select a route, complete post-screen ratings, then continue to the other interface according to the counterbalanced order. In the baseline condition, participants choose among complete route alternatives. In the prototype condition, participants choose a main route and then

choose compatible first-mile and last-mile connectors, allowing non-walk options to be selected when they better match the participant's needs.

Q1: ฉันเข้าใจข้อมูลการเดินทางบนหน้าจอได้ง่าย I could understand the route information on this screen

Clarity / understandability

(Clarity/understandability) – 1–5

1

2

3

4

5

ไม่เห็นด้วยอย่างยิ่ง

ไม่เห็นด้วย

ปานกลาง

เห็นด้วย

เห็นด้วยอย่างยิ่ง

Q2: หน้าจอนี้ช่วยให้ฉันเปรียบเทียบตัวเลือกและตัดสินใจเลือกเส้นทางได้ง่าย This screen helped me compare options and make a route decision

Ease of decision-making

(Ease of decision-making) – 1–5

1

2

3

4

5

ไม่เห็นด้วยอย่างยิ่ง

ไม่เห็นด้วย

ปานกลาง

เห็นด้วย

เห็นด้วยอย่างยิ่ง

Q3: ฉันบอกได้ชัดเจนว่าอะไรคือช่วงต้นทาง ช่วงหลัก และช่วงปลายทางของเส้นทางที่เลือก I clearly understood the first-mile / main transit / last-mile segments of the chosen route (FMLM understanding) – 1–5

First-mile / main-transit / last-mile understanding

1

2

3

4

5

ไม่เห็นด้วยอย่างยิ่ง

ไม่เห็นด้วย

ปานกลาง

เห็นด้วย

เห็นด้วยอย่างยิ่ง

Q4: เส้นทางที่ฉันเลือกตรงกับสิ่งที่ฉันต้องการสำหรับสถานการณ์นี้ The routes shown matched what I actually need for this scenario

Subjective preference fit

(Subjective preference fit) – 1–5

1

2

3

4

5

ไม่เห็นด้วยอย่างยิ่ง

ไม่เห็นด้วย

ปานกลาง

เห็นด้วย

เห็นด้วยอย่างยิ่ง

บันทึกและไปต่อ

Figure 8: Screen 3-7 of the system, end-of-interface survey. Note that this is directly below Figure 1 and 2 in the system and coexists within the same screen.

- Final questionnaire, as shown in Figure 9. After completing both conditions, participants compare App A and App B overall and rate the usefulness of the FMLM decomposition and connector-selection feature.

ความคิดเห็นโดยรวมหลังการทดลอง Overall Opinion

หลังจากได้ทดลองใช้งานทั้งสองรูปแบบ กรุณาเปรียบเทียบประสบการณ์โดยรวมของคุณ

(Scale: 1 = App A much better → 3 = Equal → 5 = App B much better)

1 = แอป A ดีกว่ามาก · 2 = แอป A ดีกว่าเล็กน้อย · 3 = พอดีกัน · 4 = แอป B ดีกว่าเล็กน้อย · 5 = แอป B ดีกว่ามาก (Scale: 1 = App A much better → 3 = Equal → 5 = App B much better)

Overall decision support

Q2: แอปใดช่วยให้คุณเลือกเส้นทางที่ตรงกับความต้องการของคุณมากกว่า Which app helped you find routes that better matched your needs?

Preference alignment

Q3: แอปใดทำให้คุณเข้าใจตัวเลือกการเดินทางได้ชัดเจนกว่า Which app made route selection clearer?

Overall clarity

Q4: หากต้องใช้งานจริงในชีวิตประจำวัน คุณอยากใช้แอปใดมากกว่า — If you needed this in real daily life, which app would you prefer?

Behavioral intention to use

1. แอป A ดีกว่ามาก
2. แอป A ดีกว่าเล็กน้อย
3. พอ ๆ กัน
4. แอป B ดีกว่าเล็กน้อย
5. แอป B ดีกว่ามาก

Q5: การแสดงข้อมูลช่วงต้นทาง-ช่วงหลัก-ช่วงปลายทาง (First Mile / Main Transit / Last Mile) มีประโยชน์ต่อคุณเพียงใด

Perceived usefulness of EMI-M visualization

How useful was showing First Mile / Main Transit / Last Mile segments separately? (Perceived usefulness of FMLM visualization)

1 2 3 4 5

ไม่มีประโยชน์เลย ไม่ค่อยมีประโยชน์ ปานกลาง ค่อนข้างมีประโยชน์ มีประโยชน์มาก

– 1 (Not useful at all) to 5 (Very useful)

– 1 (Not useful at all) to 5 (Very useful)

Q6: ความคิดเห็นเพิ่มเติมเกี่ยวกับระบบนี้ (ไม่บังคับ)

เช่น สิ่งที่ชอบ สิ่งที่เราปรับปรุง หรือข้อเสนอแนะอื่น ๆ

Any additional comments about the system?

(Open text – e.g. likes, suggested improvements, other recommendations)

ส่งคำตอบสุดท้าย

6. Thank-you screen, as shown in Figure 10. Participants see a Thai completion message thanking them for their contribution to the research.



Figure 10: Final screen of the system, a thank you message

3.2.6 Data Logging

The system records participant ID, condition order, background responses, preference ranking, selected routes, selected first-mile/main/last-mile components, total time, total cost, walking/distance information, per-interface ratings, final questionnaire responses, and interaction events.

3.3 Evaluation Metrics, Analyses, and Hypotheses

As mentioned before, this study aims to evaluate the prototype compared to the baseline. Here, we will use one primary metric and few secondary measures. Then, we will explore how the analyses, both quantitative and qualitative, shall be done, and lastly, covers the study's hypotheses.

3.3.1 Primary Metric

The primary metric is preference-aligned route choice. This metric measures how closely each participant's selected route matches the preference ranking they provided in the background questionnaire. For example, if a participant ranked cost as their most important factor, lower-cost route combinations receive higher alignment scores. Similarly, if walking distance was ranked as most important, route combinations with shorter first-mile and last-mile walking distances receive higher scores.

The scoring procedure is implemented using the scoring version `preference_fit_v1`. For each scenario, the system defines a set of candidate routes. In the baseline condition, the interface-relative candidate set contains each main route paired with its default walking first-mile and last-mile connectors. In the prototype condition, the interface-relative candidate set contains all compatible combinations of first-mile connector, main route, and last-mile connector that participants could construct through the prototype interface. In addition, a candidate set for a common universe is defined. This set contains all compatible route combinations within the scenario and is used as a complementary comparison across both interface conditions.

Following multi-criteria personalized route-planning approaches, candidate routes are evaluated by normalizing route attributes and aggregating them using preference-based weights [36, 37]. For each candidate route, five attributes are considered: total travel time, total cost, first-mile and last-mile walking distance, reliability, and simplicity. Travel time, cost, distance, and simplicity are normalized within the relevant candidate set, since lower values are considered better. The normalized score is calculated as follows:

$$N_x(c) = 1 - (x_c - \min(x)) / (\max(x) - \min(x)) \quad (2)$$

where x represents travel time, cost, distance, or simplicity across all candidates in the candidate set. If all candidates have the same value for a given attribute, the normalized score for that attribute is set to 1 for all candidates.

Reliability is mapped directly onto a 0-1 scale. Values for high, medium, and low reliability are assigned as 1.0, 0.6, and 0.2, respectively. Simplicity cost is defined as the number of transfers, plus the number of main-route modes, plus the number of connector legs.

Participant preference weights are derived from the drag-ranked preference question in the background questionnaire. This ranking-to-weight conversion is a study-specific operationalization of the preference-weighting step used in multi-criteria route evaluation. The highest-ranked factor receives 5 points, the second receives 4 points, and this continues until the lowest-ranked factor receives 1 point. These points are then normalized so that all five weights sum to 1:

$$w_j = p_j / \sum(p_j), \text{ where } p_j = 6 - \text{rank}_j \quad (3)$$

The weighted preference score for each candidate route is then calculated using the normalized weighted-sum logic common in multi-criteria route evaluation [36, 37]:

$$S(c) = w_{\text{time}}N_T(c) + w_{\text{cost}}N_K(c) + w_{\text{walk}}N_D(c) + w_{\text{reliability}}R_c + w_{\text{simplicity}}N_U(c) \quad (4)$$

The selected route is converted into a preference-fit ratio by dividing its score by the best possible score in the relevant candidate set. This ratio form is a study-specific normalization used to express how close the selected route is to the best-scoring available alternative under the participant's stated preferences. Two variants are reported. The first is the interface-relative preference-fit ratio, which compares the selected route against the best route available within the same interface condition:

$$\text{Interface Relative Preference Fit Ratio} = S(c_{\text{selected}}) / \max(S(c)), \text{ for all } c \text{ in } C_{\text{interface}} \quad (5)$$

The second is the common-universe preference-fit ratio, which compares the selected route against the best route available across all compatible route combinations in the scenario:

$$\text{Common Universe Preference Fit Ratio} = S(c_{\text{selected}}) / \max(S(c)), \text{ for all } c \text{ in } C_{\text{common}} \quad (6)$$

Both scores range from 0 to 1, with 1 indicating that the participant selected the highest-scoring route according to their stated preferences. Scores are rounded to three decimal places for storage and reporting.

Preference alignment is calculated separately for the baseline and prototype conditions. This allows us to evaluate whether the prototype helps participants select complete route combinations that better match their stated priorities, rather than assuming that there is a single objectively correct route for all users.

3.3.2 Secondary Measures

Secondary measures include per-scenario clarity, decision ease, FMLM understanding, preference fit, final comparative preference, perceived usefulness of FMLM information and connector choice, intention to use, route choice patterns by condition, and optional qualitative feedback.

3.3.3 Quantitative Analysis

Here we compare baseline and prototype outcomes within participants. Preference-aligned route scores and per-scenario Likert ratings are compared across conditions using the Wilcoxon signed-rank test, a rank-based non-parametric test for paired observations commonly used when normality of paired differences should not be assumed [49, 50]. Final comparative ratings are analyzed descriptively and against the neutral midpoint. Additional exploratory analyses examine whether results differ by counterbalanced condition order, scenario, participant priority subgroup, participant background group, and scenario position within the task sequence. Effect sizes are reported as paired standardized mean differences where appropriate.

Graphs and result tables are generated in the admin results dashboard, including baseline-versus-prototype comparisons for route scores, clarity, decision ease, FMLM understanding, preference fit, final preference, and usefulness of FMLM information, as shown in Figure 11. Figures are exportable for use in the paper.

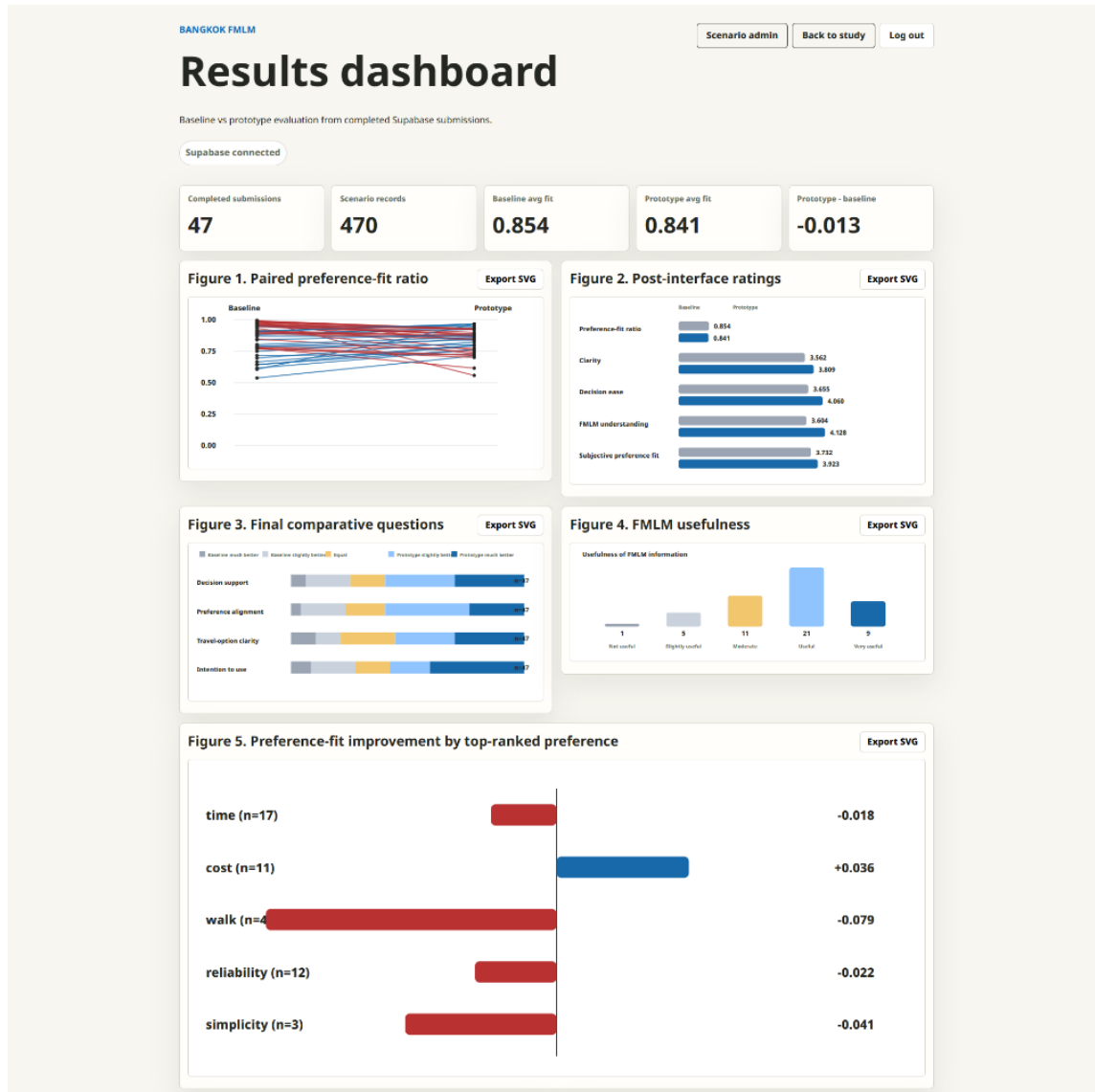


Figure 11: The results dashboard within the system inside the admin screen.

Incomplete responses are excluded from the analysis if participants do not finish all assigned scenarios under both conditions. Partial logs may be retained for debugging or exploratory dropout analysis.

3.3.4 Qualitative Analysis

Open-ended feedback is analyzed thematically to identify recurring comments about route comprehension, decision difficulty, interface clarity, perceived usefulness of FMLM decomposition and connector selection, missing information, or confusion. Information here will be used to provide context towards the quantitative results and design implications.

3.3.5 Hypotheses

Therefore, our study will propose three research hypotheses to be considered in the next section as follows:

7. H1: Explicit presentation and selectable choice of first- and last-mile access options improves users' ability to select routes aligned with their travel preferences.
8. H2: The prototype improves users' perceived understanding of travel trade-offs compared to the baseline interface.
9. H3: The prototype increases user awareness and use of alternative FMLM access modes beyond walking when connecting to reliable transit.

4 RESULTS

4.1 Dataset Integrity

From the 46 completed submissions, all contained complete background responses, preference rankings, selected routes, per-interface ratings, final comparative questionnaire responses, and FMLM usefulness ratings. No missing route selections, missing preference-fit scores, missing post-screen ratings, invalid 1-5 scale values, or incomplete baseline/prototype scenario pairs were found.

4.2 Quantitative Results

The main calculated preference-fit analysis used the common-universe metric, which compares selections from both interfaces against the best route combination available in the full scenario candidate set. Under this metric, the average preference-fit ratio was nearly the same in the two conditions, with the baseline slightly higher ($M = 0.843$) than the prototype ($M = 0.839$), a small difference of -0.005 . A stored interface-relative sensitivity check, which compares each selection only against the best option available within that interface's own candidate set, showed a slightly larger baseline-favoring difference (baseline $M = 0.852$, prototype $M = 0.839$, difference = -0.013). This indicates that the prototype did not produce a full-sample improvement in calculated route-choice quality, but the size of the disadvantage was small and depended on the scoring denominator.

At the participant level, the common-universe metric was slightly positive for more participants than it was negative. We found 23 participants had higher calculated preference-fit scores with the prototype, 17 had higher scores with the baseline, and 6 were tied. Therefore, the slightly baseline-favoring mean should not be interpreted as a uniform baseline advantage. Instead, it indicates that the prototype helped more participants by this metric, but a smaller number of larger negative differences in specific participants or scenarios pulled the overall mean slightly below the baseline.

Participant-level common-universe preference-fit differences

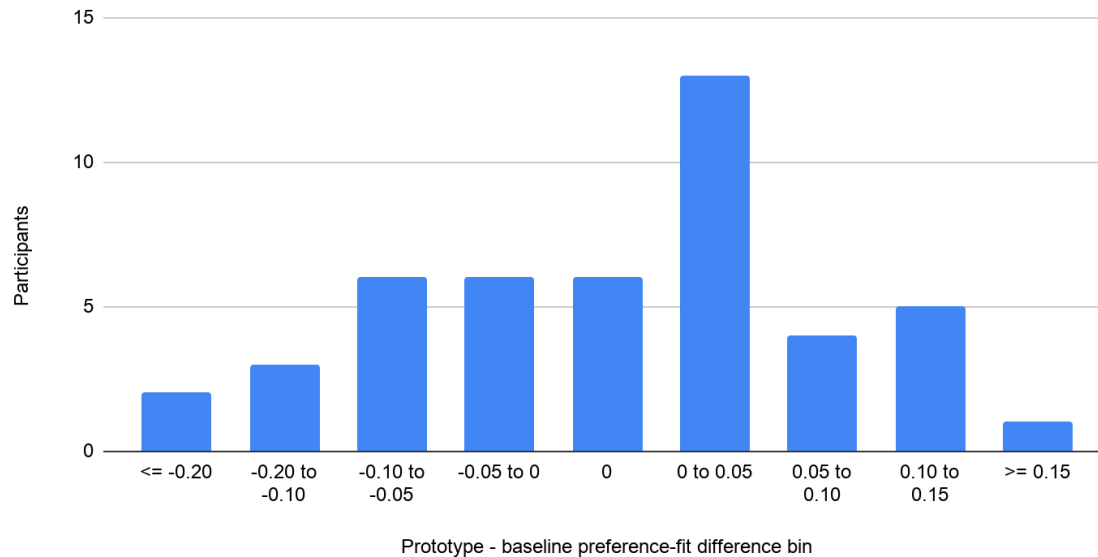


Figure 12: Participant-level distribution of common-universe preference-fit differences

Figure 12 visualizes this participant-level spread, making clear that the near-zero average difference reflects mixed participant-level effects rather than consistent superiority of either interface. In contrast, subjective post-interface ratings in Figure 13 favored the prototype. Compared with the baseline, the prototype received higher average ratings for clarity, ease of decision-making, FMLM understanding, and self-reported preference fit, as shown in Table 4:

Table 4: Main quantitative results by condition

Measure	Baseline mean	Prototype mean	Difference	Dz	p
Common-universe preference-fit ratio	0.843	0.839	-0.005	-0.04	.672
Clarity	3.543	3.787	+0.243	0.26	.099
Decision ease	3.648	4.043	+0.396	0.44	.005
FMLM understanding	3.587	4.113	+0.526	0.63	<.001
Self-reported preference fit	3.726	3.904	+0.178	0.26	.077

Preference-fit ratio ranges from 0 to 1, where higher values indicate a closer match between the selected route and the participant's stated preference ranking. Self-reported rating means are on a 1-5 scale. Difference is calculated as prototype minus baseline, so positive values favor the prototype. dz refers to Cohen's dz, an effect size for paired within-subject comparisons. p is the Wilcoxon signed-rank test p-value; p < .05 is treated as statistically significant for this analysis.

The common-universe preference-fit ratio showed a near-zero effect (mean difference = -0.005, dz = -0.04, p = .672), indicating no clear calculated route-choice advantage for either interface. In contrast, subjective ratings favored the prototype, with the largest effect for FMLM understanding (+0.526, dz = 0.63, p < .001), followed by decision ease (+0.396, dz = 0.44, p = .005). Clarity and self-reported preference fit showed smaller positive effects. This pattern suggests

that the prototype improved perceived comprehension and decision support more clearly than calculated preference-aligned route choice.

Post-interface ratings by condition

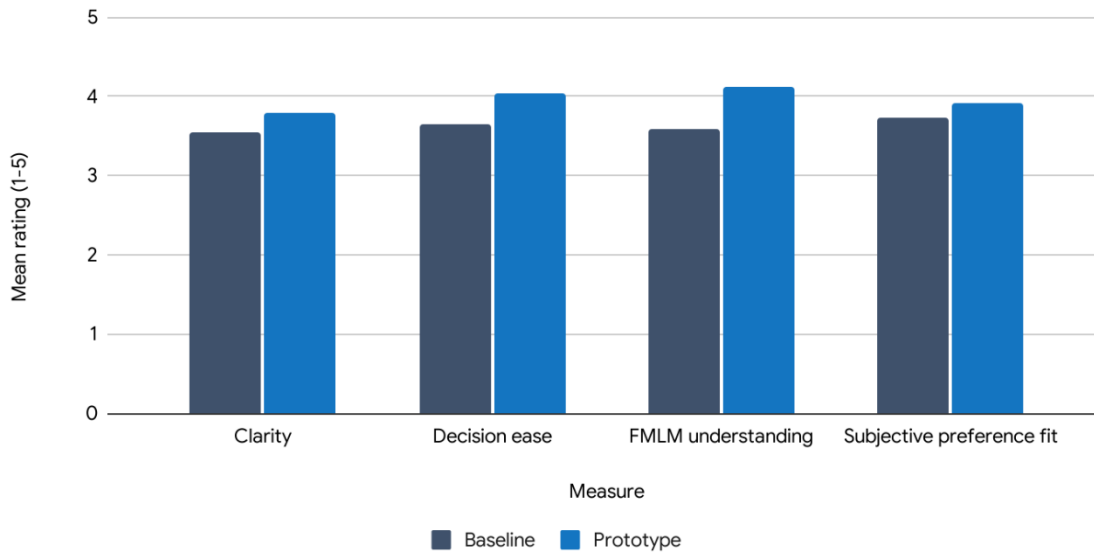


Figure 13: Mean post-interface ratings by condition.

4.2.1 Scenario-Level Pattern

Scenario-level results were mixed under both calculated metrics shown in Figure 14. Using the common-universe metric, the prototype showed its clearest improvements in the Bang Na to Sathon scenario (+0.119), followed by Suan Luang to Ekkamai (+0.023), Bang Wa to Chula/Siam (+0.014), and Min Buri to Asok (+0.012). The largest negative differences appeared in Phra Khanong to Grand Palace (-0.106), Bang Phlat to Ari (-0.071), and Lak Si to Silom (-0.065). This indicates that the prototype's benefit is scenario-dependent and may rely on whether the available FMLM alternatives are realistic, understandable, and meaningfully different from the baseline walking connectors.

Scenario-level preference-fit difference

Prototype minus baseline; positive values favor prototype

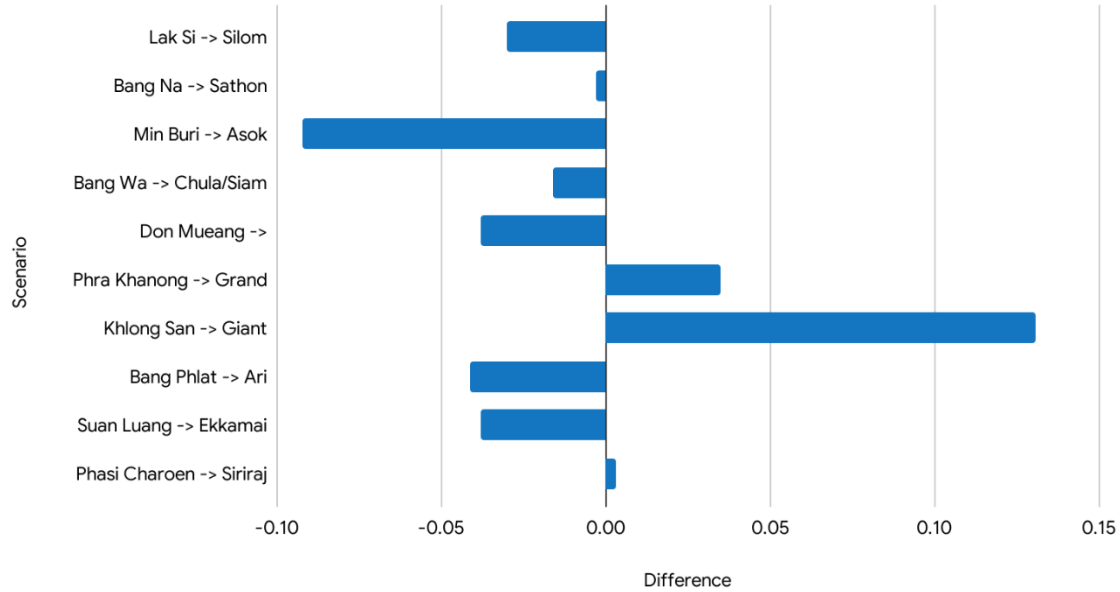


Figure 14: Scenario-level common-universe prototype-minus-baseline preference-fit differences.

4.2.2 Order-Effect Verification

Condition order was balanced in the completed dataset: 23 participants experienced the prototype first and 23 experienced the baseline first. The prototype advantage in subjective ratings was present in both order groups, but it was stronger among participants who saw the prototype first. In the prototype-first group, the prototype-baseline differences were +0.357 for clarity, +0.470 for decision ease, +0.696 for FMLM understanding, and +0.365 for self-reported preference fit. In the baseline-first group, the corresponding differences were +0.130, +0.322, +0.357, and -0.009. This pattern suggests that order and first impression may affect subjective experience, especially because the prototype has a higher learning burden. However, both groups still rated the prototype higher for decision ease and FMLM understanding.

The stored interface-relative calculated preference-fit ratio did not show the same order pattern. The prototype-first group had a near-zero difference (+0.002), while the baseline-first group slightly favored baseline (-0.027). This reinforces the interpretation that participants perceived the prototype as more informative, but the current interface did not reliably convert that perception into higher preference-aligned route choices.

4.2.3 Participant Priority Subgroups

Exploratory subgroup analysis by each participant's top-ranked route priority showed that the prototype's benefit varied by preference type under the stored interface-relative metric. Participants who prioritized cost showed a small positive preference-fit ratio difference for the prototype (+0.042, $n = 10$). Participants prioritizing time (-0.018, $n = 17$), reliability

(-0.022, $n = 12$), simplicity (-0.041, $n = 3$), and walking minimization (-0.079, $n = 4$) did not show a calculated preference-fit advantage for the prototype. Because some subgroups are small, these patterns should be treated as descriptive rather than conclusive.

Subjective ratings were more consistently positive. Decision ease improved for the time (+0.482), reliability (+0.350), simplicity (+0.733), and cost (+0.440) subgroups, while the walking-minimization subgroup showed a negative difference (-0.200). This suggests that the prototype may be more effective for comparing trade-offs such as time, cost, and reliability than for participants whose dominant concern is minimizing connector distance.

4.2.4 Participant Background Subgroups

Additional descriptive subgroup analysis examined whether participant background affected the prototype-baseline difference. These analyses are exploratory because the study was not powered to test group-by-interface interactions.

Travel frequency showed a larger subjective prototype advantage among less frequent travelers than frequent travelers. Less frequent travelers ($n = 11$) showed prototype-baseline differences of +0.582 for clarity, +0.673 for decision ease, +0.618 for FMLM understanding, and +0.382 for self-reported preference fit. Frequent travelers ($n = 35$) still favored the prototype subjectively, but with smaller differences: +0.137 for clarity, +0.309 for decision ease, +0.497 for FMLM understanding, and +0.114 for self-reported preference fit. This suggests the prototype may be especially helpful for people who travel less routinely and may need more support interpreting unfamiliar route structures.

Navigation-app usage showed a similar pattern. Participants who rarely or never used navigation apps ($n = 10$) showed the largest prototype advantage: +0.580 for clarity, +0.760 for decision ease, +0.820 for FMLM understanding, and +0.360 for self-reported preference fit. Occasional navigation-app users ($n = 28$) showed smaller but still positive differences, while heavy navigation-app users ($n = 8$) showed a strong gain mainly in FMLM understanding (+0.675). This suggests that the prototype's decomposed route information may be most valuable for users with less existing navigation-tool fluency.

Familiarity with multi-stage travel also shaped the results. Participants with low multimodal familiarity ($n = 15$) showed relatively strong subjective gains from the prototype: +0.440 for clarity, +0.693 for decision ease, +0.733 for FMLM understanding, and +0.280 for self-reported preference fit. Participants with high multimodal familiarity ($n = 19$) also favored the prototype, but less strongly: +0.221 for clarity, +0.400 for decision ease, +0.537 for FMLM understanding, and +0.211 for self-reported preference fit. The moderate-familiarity group ($n = 12$) showed weaker subjective differences. These results are consistent with the qualitative feedback that first-mile and last-mile information is especially useful when users are less familiar with a route or with complex multimodal travel.

Subgroup patterns suggest that the prototype may be most beneficial for less frequent travelers and participants with lower navigation-app experience. These groups rated the prototype more positively and, under the stored interface-relative metric, showed small positive differences in calculated preference-fit ratio. However, the pattern was not consistent across all participant backgrounds, and the subgroup sizes were small and not balanced by design. Therefore, these findings should be interpreted as exploratory evidence that FMLM visualization may be especially useful for less routine or less navigation-fluent travelers, rather than as conclusive proof of improved route-choice quality for every group.

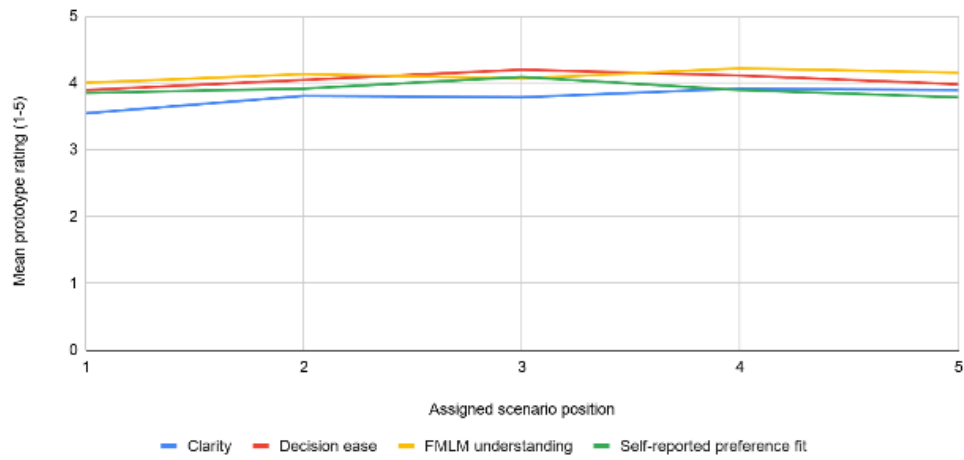
4.2.5 Learning Across Multiple Scenarios

Prototype ratings by scenario position indicate a modest learning effect as shown in Figure 15. Prototype clarity increased from 3.54 in the first assigned scenario to 3.80 in the second scenario, 3.78 in the third, 3.91 in the fourth, and 3.89 in the

fifth. Prototype FMLM understanding similarly increased from 4.00 in the first scenario to 4.15 in the fifth. This supports participant comments that the prototype was initially harder to understand but became more usable after repeated exposure.

However, calculated preference-fit did not show the same learning pattern. Under the common-universe metric, baseline preference-fit ratios by assigned scenario position were 0.833, 0.816, 0.860, 0.870, and 0.833, while prototype ratios were 0.824, 0.829, 0.854, 0.868, and 0.808. The corresponding prototype-minus-baseline differences were -0.009, +0.013, -0.007, -0.003, and -0.024 for positions 1-5. These values fluctuate around zero rather than improving steadily across repeated exposure. Therefore, repeated use appears to improve perceived comprehension, while showing no clear corresponding improvement in calculated preference-aligned route choice.

Prototype subjective ratings by scenario position



Common-universe preference-fit by scenario position

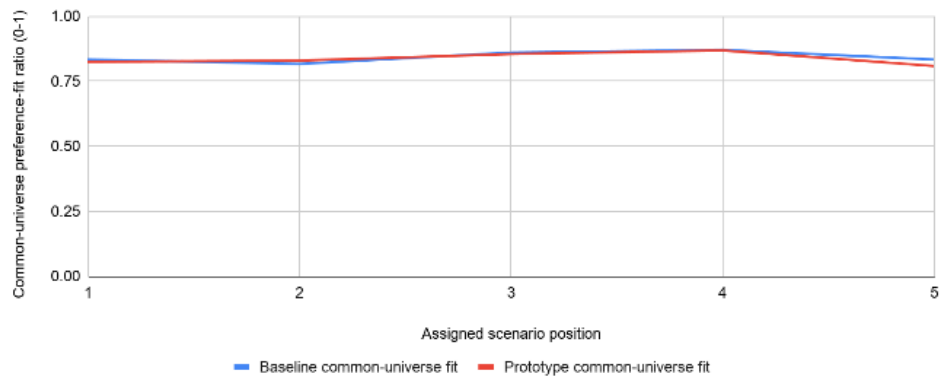


Figure 15: Prototype subjective ratings and Common-universe preference-fit by scenario position

4.2.6 Final Questionnaire Comparative Ratings

Final comparative ratings also leaned toward the prototype as shown in Table 5 and Figure 16. On the 5-point comparative scale, where 1 favored the baseline strongly, 3 indicated equality, and 5 favored the prototype strongly, the prototype side received more responses than the baseline side across all four final comparative dimensions:

Table 5: Final comparative questionnaire results

Measure	Baseline preferred	Equal	Prototype preferred	Mean
Decision support	12	7	27	3.543
Preference alignment	11	8	27	3.522
Overall clarity	10	11	25	3.522
Intention to use	13	7	26	3.587

The 5-point scale ranges from baseline strongly preferred to prototype strongly preferred; the three count columns group responses into baseline-side, midpoint, and prototype-side choices.

The separate FMLM usefulness item had a mean score of 3.674 out of 5. Most participants rated the FMLM information as moderately to highly useful, with 29 of 46 responses falling on the positive side of the scale.

Final comparative preferences

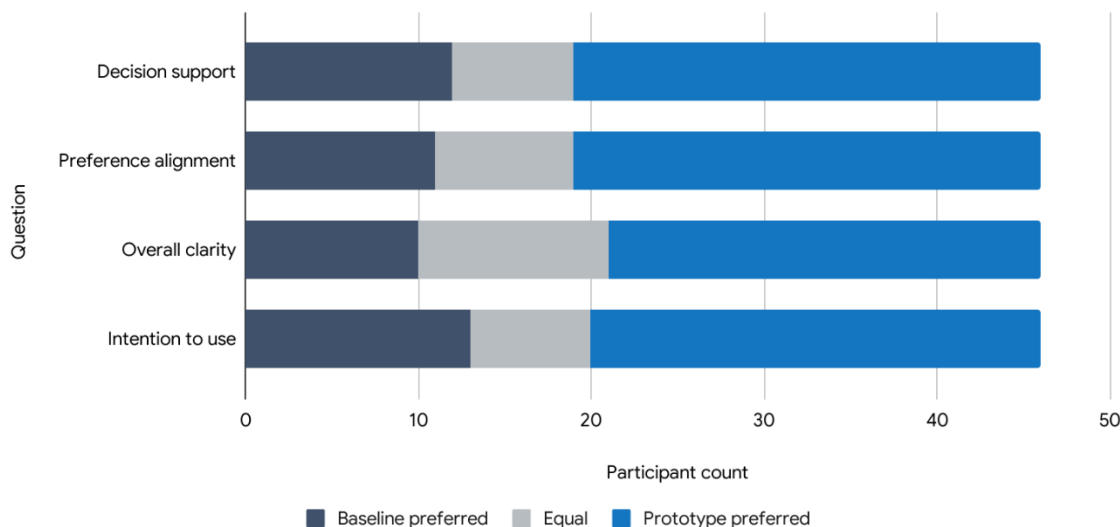


Figure 16: Final comparative preferences across four dimensions

4.3 Qualitative Feedback

Nine participants provided open-ended comments. These comments help explain why subjective ratings favored the prototype while the calculated preference-fit scores did not clearly improve. Thematic coding of the nine optional comments produced six recurring themes. Because a single comment could contain multiple themes, counts should be read as theme occurrences rather than mutually exclusive categories: richer information and customization (5 comments), usability complexity or learning burden (4 comments), map or visual encoding confusion (3 comments), route-data realism concerns (2 comments), usefulness for unfamiliar routes or first-time route users (1 comment), and terminology confusion around First Mile / Last Mile language (1 comment).

First, participants valued the prototype's richer information and customization. Comments described App B as useful because it provided more information about time, cost, distance, reliability, route structure, and alternative route components. Participants also appreciated that the prototype did not force a single predetermined route and allowed them to adjust the route based on their own needs.

Second, participants saw value in first- and last-mile information, especially for unfamiliar trips. One participant noted that First Mile / Last Mile information was useful for everyone and especially helpful for people who had not traveled on that route before. This suggests that the concept may be particularly valuable for unfamiliar areas, visitors, newcomers, or

users planning routes outside their normal routine, although first-time city users were not directly tested as a separate group.

Third, the prototype had a clear usability cost. Several comments described App B as more detailed but also harder to understand at first. Participants mentioned needing to move back and forth between route sections, compare many options, and spend time learning how the interface worked. One comment described App A as concise and easy to understand immediately, while App B initially felt visually overwhelming but became more useful after the participant understood the logic.

Fourth, map and visual encoding issues affected comprehension. Participants reported confusion about origin and destination markers, the meaning of red and green points, route geometry, and whether some map paths reflected realistic walking or driving routes. These issues may have reduced trust in some route recommendations and weakened the prototype's advantage.

Fifth, terminology was a barrier. A participant suggested avoiding technical labels such as First Mile and Last Mile and using simpler everyday labels. Future interfaces should describe the same concept in participant-friendly terms, such as first step from origin to main travel segment (ต้นทาง), main travel segment (เส้นทางหลัก), and last step from main travel segment to destination (ปลายทาง). This supports the interpretation that the prototype's underlying concept was useful, but its current language was not always participant friendly.

Sixth, some route options appeared unrealistic to participants. One comment criticized routes that seemed indirect or unreasonable compared with simply taking Grab. This suggests that route-data realism is an important limitation: if participants do not trust the available route alternatives, the interface may be judged negatively regardless of its visualization concept.

4.4 Interpretation

A key result is the split between calculated preference-fit and perceived usefulness. The main common-universe preference-fit ratio was close to parity but slightly favored the baseline (0.843 vs. 0.839), while the stored interface-relative sensitivity check showed a larger baseline-favoring difference (0.852 vs. 0.839). In contrast, participants rated the prototype higher for clarity, decision ease, FMLM understanding, self-reported preference fit, final comparative preference, and usefulness of first- and last-mile information. Overall, the results suggest that the prototype improved participants' perceived understanding of trip structure and their subjective ability to compare route options. The strongest evidence is found in the subjective ratings and final comparative questionnaire, where participants consistently leaned toward the prototype. The open-ended comments also support the value of explicit first- and last-mile information, particularly for unfamiliar trips. Therefore, the prototype's main contribution in the current study is better understood as improved comprehension and decision support rather than a demonstrated improvement in calculated route-choice optimality.

However, the prototype did not improve calculated preference-fit in the full-sample result. The main common-universe metric was close to parity but slightly favored the baseline, and the stored interface-relative sensitivity metric showed a somewhat larger baseline-favoring gap. This should be interpreted carefully because preference-fit is a structured proxy based on normalized time, cost, distance, reliability, and simplicity; it may not fully capture route trust, comfort, perceived realism, safety, weather exposure, transfer annoyance, or the cognitive effort required to compare many options. The findings therefore support the value of FMLM-aware route planning as a concept, while also showing that the current prototype needs clearer information hierarchy, less technical terminology, improved map encoding, and more realistic route options before its perceived benefits can reliably translate into higher calculated preference-aligned route scores.

5 DISCUSSION

The central finding of this study, that the prototype improved subjective experience without producing a corresponding improvement in calculated preference-fit, is consistent with the distinction drawn in prior literature between information availability and preference-fit. As noted by [26], commuters require complete door-to-door information to feel confident in their journeys, and the absence of any segment — including where to board, how to transfer, or what connectors are available — increases perceived difficulty disproportionate to the actual travel burden. The prototype's positive gains in clarity (+0.243) and FMLM understanding (+0.526) suggest it successfully addressed this information gap. However, improved comprehension of route structure does not necessarily redirect participants toward routes that score higher on the preference-fit metric, which is based on normalized time, cost, distance, reliability, and simplicity alone. This echoes findings from [17] and [18], which show that real-world mode choice incorporates factors such as safety, comfort, and walking environment that are difficult to capture in a structured scoring formula.

The prototype's strongest subjective advantage appeared in FMLM understanding and decision ease, which aligns directly with the argument in [28] that route information structure affects cognitive load. The linear decomposition of FMLM segments in the prototype — separating first-mile connector, main transit, and last-mile connector — mirrors the finding that a clearly hierarchical, linear route map reduces cognitive load compared to a map-based equivalent. That said, qualitative feedback indicated that several participants found the prototype initially overwhelming, needing to navigate between route sections before the logic became clear. This learning curve is consistent with the order-effect analysis, which showed stronger subjective gains in the prototype-first group, and with the scenario-position data showing clarity improving from 3.54 to 3.89 across five consecutive scenarios. The interface appears to become more usable with exposure, which has practical implications for deployment.

The scenario-level heterogeneity in prototype-minus-baseline differences, ranging from +0.119 (Bang Na → Sathon) to -0.106 (Phra Khanong → Grand Palace), is interpretable considering Bangkok's specific transit geography. The prototype showed the clearest gains in scenarios where FMLM alternatives were meaningfully differentiated and realistically accessible, such as Bang Na → Sathon where the soi depth of Bangkok Patana School creates a genuine first-mile decision between walking to BTS, using a van service, or waiting for a bus along the expressway underpass. In contrast, negative differences appeared in scenarios such as Phra Khanong → Grand Palace and Bang Phlat → Ari, where FMLM alternatives may have appeared less credible or less practical than simply using Grab, consistent with the qualitative concern that some route options seemed indirect compared to a ride-hailing alternative.

This finding reinforces what [11] and [25] describe as a core challenge in Bangkok FMLM planning: the informal nature of many connector modes means their route data is incomplete, inconsistent, or geographically misleading. The songthaew dataset discussed in [12], for instance, contains outdated route paths for some lines, which could directly contribute to participant skepticism about certain connector options presented in the prototype. Until the underlying route data is complete and more trustworthy, the prototype's visualization concept may be constrained by the reliability of its inputs regardless of its design quality.

The subgroup patterns by travel frequency, navigation-app experience, and multimodal familiarity suggest the prototype's benefit is not uniform across users. Less frequent travelers, infrequent navigation-app users, and those with lower multimodal familiarity showed substantially larger subjective gains than their counterparts. This is consistent with prior work showing that public-transport users need complete pre-trip and en-route information to reduce uncertainty, especially when journeys are unfamiliar [26, 27].

The negative subjective outcome for the walking-minimization subgroup ($n = 4$, decision ease $\Delta = -0.200$) is a notable exception. A plausible explanation is that explicitly decomposing each walking segment — as the prototype does — makes

unavoidable walking distances more salient and more difficult to compare across routes, rather than less. For users whose primary concern is minimizing walking, this increased salience may function as an added burden rather than a clarification, particularly when no route option fully satisfies their preference. This parallels the finding from [17] that walking is preferred when it is simple and cost-free, but that making walking-distance trade-offs explicit may alter how commuters perceive those costs.

The qualitative finding that the term "First Mile / Last Mile" was unfamiliar or unclear to some participants connected to a broader issue in transit interface design. As [28] and [31] both suggest, the effectiveness of route information depends not only on what is presented but on how intuitively users can interpret the schema. The suggestion from one participant to replace FMLM terminology with participant-friendly Thai equivalents - *ต้นทาง* (first step from origin to main travel segment), *เส้นทางหลัก* (main travel segment), and *ปลายทาง* (last step from main segment to destination) - is practically significant and consistent with the localization requirements of any transit tool designed for Bangkok's diverse commuter population.

Taken together, these findings support the conclusion of [29] that route recommendation should be treated as a multi-attribute decision problem rather than a single shortest-path solution, and that presenting Pareto-relevant alternatives - differentiated on time, cost, walking, and reliability - produces more useful decision support than a single ranked recommendation. The prototype operationalizes this principle for Bangkok's FMLM context. However, the results also suggest that three additional conditions must be met before such an interface can reliably improve calculated route-choice quality: the underlying transit data must be complete and trustworthy enough that all presented options are genuinely feasible; the information hierarchy must be simple enough to be navigable without prior learning; and the presentation of walking segments in particular must be handled carefully to avoid increasing rather than reducing perceived decision complexity for walking-sensitive users

6 LIMITATIONS

Due to time and spatial constraints, this study focused on the top 10 routes in Bangkok identified through surveys. However, this approach presents several limitations, as the availability of transportation options in certain areas cannot be guaranteed due to factors such as vehicle frequency, service accessibility, passenger preferences, and routes commonly used by residents. As a result, the actual travel conditions may be different from the options presented in the prototype application. Therefore, the prototype primarily relies on estimated data rather than real-time conditions. Moreover, the study was conducted online, and participants completed tasks on their own devices without supervision. Differences in screen size, familiarity with Bangkok, and prior experience with multimodal transport may have affected both route choices and interface ratings.

The preference-fit metric also has construct validity limitations. The metric operationalizes route quality as a weighted sum of normalized time, cost, walking distance, reliability, and simplicity based on each participant's stated preference ranking. However, stated rankings collected before the task may not reflect the weights participants actually applied when choosing a route under realistic trade-off conditions. Factors such as perceived route safety, trust in connector data, weather exposure, and transfer annoyance are not captured in the metric, which may explain part of the gap between subjective ratings and calculated preference-fit scores.

In part of the prototype survey, some participants did not clearly understand the terms "First Mile" and "Last Mile" and found the prototype confusing at first, as it required understanding a trip as multiple separate components. A few reported that it became easier to use after trying several scenarios. This suggests that early ratings may reflect initial confusion

rather than the actual usability of the interface. For future versions, in survey form app comparisons should display on a single page for easier comparison. The next version of the app should include a more detailed quick start guide for new users and the interface could also be improved to show First-Mile / Main / Last-Mile in columns and show all methods simultaneously for easier viewing and faster decision-making.

A further limitation concerns the challenge of applying international frameworks to the Thai context. Prior work on personalized public transport recommendation in Japan highlights the value of adapting route suggestions to user preferences and interface context [32], while commercial transit planners such as Japan Transit Planner illustrate how route options can be organized around practical attributes such as time, cost, and convenience [33]. However, applying such approaches in Thailand remains challenging due to the diversity of geographical conditions and the uncertainty of available First-Mile/Last-Mile transportation options. Consequently, it is difficult to definitively determine the most convenient route for all users, as route selection may also vary depending on the user's specific context and circumstances at a given time.

For future studies, we could address these limitations in several ways. Providing a structured onboarding tutorial before testing would help separate first-use confusion from genuine usability issues. Expanding the route dataset beyond top 10 routes would also give a more representative picture of travel behavior across Bangkok. Additionally, testing the interface with a more diverse group of users, including those less familiar with public transit, would help assess how broadly the findings apply. Field-based study where participants use the app during real trips would further strengthen the understanding of how First-Mile and Last Mile (FMLM) decision support performs outside of a controlled setting.

7 CONCLUSION

This study evaluated whether explicitly presenting first-mile and last-mile access options can improve route-planning decisions in Bangkok. Using a within-subjects online experiment with 46 completed submissions, participants compared a baseline route-planning interface with a prototype that decomposed each trip into origin-side access, main transit, and destination-side access segments, and allowed first-mile and last-mile connectors to be selected separately.

The findings show that the prototype improved participants' perceived ability to understand and compare route options. Participants rated the prototype higher for clarity, decision ease, FMLM understanding, and self-reported preference fit, and final comparative ratings generally favored the prototype. Qualitative feedback further indicated that participants valued richer route information, connector choice, and the ability to inspect where different parts of a trip occur.

At the same time, calculated preference-aligned route choice did not clearly improve at the full-sample level. The common-universe preference-fit ratio was close to parity and slightly favored the baseline, while participant-level results were mixed. This indicates that the prototype's main demonstrated benefit was not a uniform improvement in calculated route optimality, but rather improved comprehension, transparency, and perceived decision support. The gap between subjective usefulness and calculated preference-fit highlights the difficulty of converting richer information into better choices when the interface is cognitively demanding.

The study therefore suggests that FMLM-aware route planning is a promising direction for Bangkok, but the design must be simplified. Future systems should preserve the ability to understand and adjust first-mile and last-mile connectors while reducing initial complexity, using clearer everyday terminology, stronger visual hierarchy, more reliable map encoding, and realistic route data. A likely direction is to first present complete route alternatives, then let users inspect and modify access and egress segments after selecting a main route.

In conclusion, explicit first-mile and last-mile visualization can help users better understand complex multimodal trips, especially in a city like Bangkok where access to main transit is often uneven. However, the value of this approach depends on interface clarity. The contribution of this study is therefore twofold: it shows that users perceive FMLM decomposition

as useful for understanding route structure, and it identifies the design challenges that must be solved before such decomposition can consistently improve preference-aligned route choice.

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